

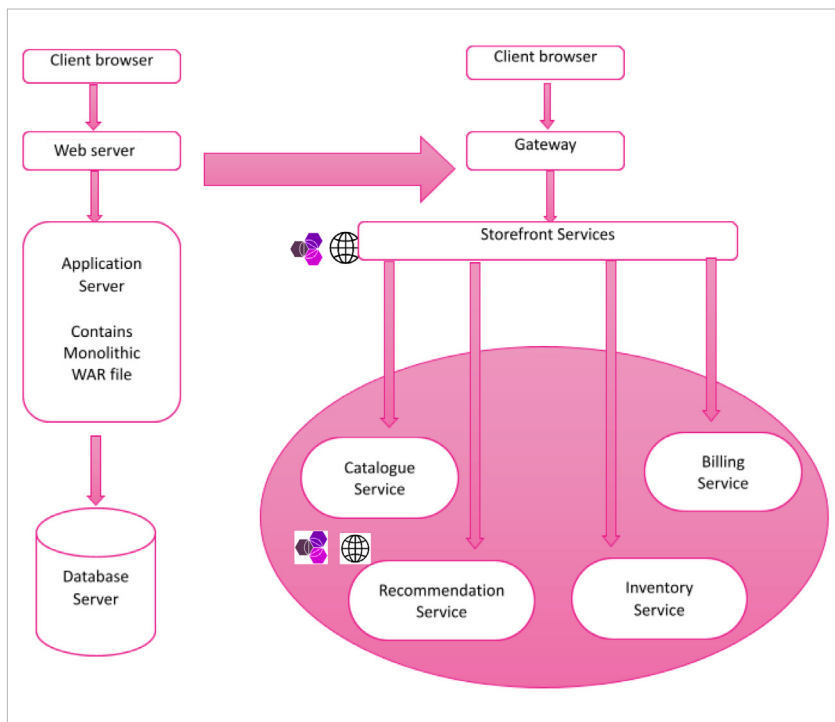


Figure 5: Flowchart of IBM Bluemix

A permissioned system is linked to a strong identity management system, with users and validators assigned to different roles. New strings of codes (chain codes) are deployed and invoked during transactions that are processed by validators through consensus. Whenever a new transaction request is initiated according to the endorsement policy, a block of data gets transferred to the committer who validates the data and checks for conflicting interests. The transaction is then committed to the ledger once it is approved by both sides. The ledger indicates the net total transactions and the hash — the global state.

Blockchain-as-a-service (BaaS) provides a platform for enterprise blockchain such as Hyperledger. Thus, a customer with access can easily create, deploy and manage an enterprise blockchain on IBM Cloud. The application will create a block and store it in the blockchain by

generating a chaincode. The block is then updated and stored in a ledger.

The following features — pluggable consensus, permissioned architecture, high modularity, open source, flexibility towards implementation, low latency of confirmation, queryable data (key-based queries and JSON queries), and support for EVM and Solidity further empower

Hyperledger Fabric. To add to this, the community of world class technology providers, individual collaborators and researchers all over the globe contribute to the diverse ecosystem by providing additional support for multiple languages like Go, Java and JavaScript.

IBM Bluemix is the IBM open cloud platform that offers platform-as-a-service (PaaS) access to Web developers and mobile developers through Cloud Foundry open technology on SoftLayer infrastructure. The cloud architecture is scalable on demand subject to workloads, and the storage could be public, private or hybrid. Bluemix supports Java, Node.js, Go, PHP, Swift, Python, Ruby on Rails, and can be extended as a support language through Buildpack (compile your app for launch) usage. Figure 5 illustrates the flow of IBM Bluemix.

Using software-as-a-service, deploying a blockchain via Bluemix provides benefits in terms of building, operating, governing and growing a solution on a multi cloud, hybrid cloud or on-premise environment. Single node deployment gives better granular control. Users have access to their private keys and Visual Studio IDE supports the development of smart contracts. **END** 🐼

References

- [1] <https://developer.ibm.com/articles/top-technical-advantages-of-hyperledger-fabric-for-blockchain-networks/>
- [2] <https://www.hyperledger.org/use/vendor-directory>
- [3] <https://www.ibm.com/topics/hyperledger>
- [4] <https://www.corda.net/>
- [5] <https://www.ibm.com/support/pages/overview-ibm-bluemix>
- [6] https://www2.deloitte.com/content/dam/insights/us/articles/6608_2020-global-blockchain-survey/DL_CIR%202020%20global%20blockchain%20survey.pdf
- [7] 51 Critical Blockchain Statistics: 2021 Data Analysis & Market Share; Financesonline.com
- [8] Key concepts of Blockchain | Blockchain Technology Fundamentals (witscad.com)

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